MP62 AM5 CPU BOARD rev A—am5 DRAFT — BLOCK DIAGRAM. Sheets are stubs (labels only). Same label name = same net (bundle).

Off-board: J1 Mini Cool Edge 224 (card edge) | J3 MCIO 124 -> base board | LUG1/LUG2 12 V | J5 debug | J4 TPM. AMD pin names (NDA) replaced by functional names.

CPU: Socket AM5 (Ryzen 7000/9000, 142 W PPT cap)

VDDCR
VDDCR_SOC
VDD_MISC
VDDIO_MEM_S3
VDD_18_S5
VDD_33_S5
VDD_MISC_S5
SVI3_CLK
SVI3_DAT
SVI3_ALERT_L
PROCHOT_L
DDR5_CHA
DDR5_CHB
SMB0_SCL
SMB0_SDA
FCH_SPI
PCIE_GFX_L0_7
PCIE_GFX_L8_15
PCIE_SSDA_LINK0_X4
PCIE_SSDA_LINK1_X4
PCIE_PROM21_X4
GPP_CLK_GFX
GPP_CLK_SSDA0
GPP_CLK_SSDA1
GPP_CLK_PROM21
PCIE_RST_L
USB10G_CPU0_3
USB2_CPU0
CLK_48M
RTC_32K
VBAT_RTC_CB
PWR_BTN_L
SYS_RESET_L
RSMRST_L
PWR_GOOD
WAKE_L
SLP_S3_L
SLP_S5_L
THERMTRIP_L
FCH_UART0

File: cpu_am5.kicad_sch

SVI3 VRM: TI TPS53685 (VDDCR + VDDCR_SOC + VDD_MISC)

12V_CB
VR_EN
VR_PGOOD
SVI3_CLK
SVI3_DAT
SVI3_ALERT_L
PROCHOT_L
VDDCRD
VDDCR_SOCD
VDD_MISCD
I2C0_SCL
I2C0_SDA

File: vrm_svi3.kicad_sch

DDR5 2DPC: 4 x UDIMM vertical (back)

5V_VIN_BULK
3V3_S5
DDR5_CHA
DDR5_CHB
SMB0_SCL
SMB0_SDA

File: ddr5.kicad_sch

PROM21 chipset — PLACEHOLDER — pinout pending datasheet

GPP_CLK_PROM21
PCIE_RST_L
PROM21_RAILS
PCIE_PROM21_X4
PCIE_ASM_X4
ASM_REFCLK
ASM_PERST_L
ASM_CLKREQ_L
PCIE_I226A_X1
I226A_REFCLK
I226A_PERST_L
I226A_CLKREQ_L
PCIE_I226B_X1
I226B_REFCLK
I226B_PERST_L
I226B_CLKREQ_L
USB10G_P21_HUBUP
USB2_P21_HUB

File: prom21.kicad_sch

Platform power: 12 V entry (LUG1/LUG2), TPS259851 eFuse, aux + PROM21 rails

5V_SBY
3V3_SB
RAIL_EN_S5
RAIL_EN_S0
RAILS_PGOOD
12V_CB
5V_VIN_BULK
3V3_S5D
VDD_18_S5
VDD_33_S5
VDD_MISC_S5
VDDIO_MEM_S3
PROM21_RAILS
I2C0_SCL
I2C0_SDA
PWR_ALERT_N

File: power.kicad_sch

EC, firmware, clocks, RTC (on-board CR2032), debug

3V3_SB
PWRBTN_N
RSTBTN_N
SLP_S3_L
SLP_S5_L
PCIE_RST_L
THERMTRIP_L
VR_PGOOD
RAILS_PGOOD
CARRIER_HOT_N
FAN_TACHIN
SMB_ALERT_N
BB_PWR_BTN_N
PWR_BTN_L
SYS_RESET_L
RSMRST_L
PWR_GOODD
VR_END
RAIL_EN_S5D
RAIL_EN_S0D
SUS_S3_ND
SUS_S5_ND
RSMRST_OUT_ND
PLTRST_ND
THERMTRIP_ND
PROCHOT_L
UART0
I2C0_SCL
I2C0_SDA
FAN_PWMOUTD
FCH_SPI
FCH_UART0
CLK_48MD
RTC_32KD
VBAT_RTC_CBD
I2C_BB

File: ec_fw.kicad_sch

J1 Amphenol ME1022410103011 Mini Cool Edge 224 card edge

PCIE_GFX_L0_7
PCIE_GFX_L8_15
PCIE_SSDA_LINK0_X4
PCIE_SSDA_LINK1_X4
GPP_CLK_GFX
GPP_CLK_SSDA0
GPP_CLK_SSDA1
USB2_CPU0
SUS_S3_N
SUS_S5_N
RSMRST_OUT_N
PLTRST_N
THERMTRIP_N
PWR_ALERT_N
FAN_PWMOUT
5V_SBY
3V3_SB
PWRBTN_N
RSTBTN_N
CARRIER_HOT_N
FAN_TACHIN
UART0
I2C0_SCL
I2C0_SDA
SMB0_SCL
SMB0_SDA
WAKE_L

File: cpu_link.kicad_sch

J3 MCIO-124 to base board (rev 2 pinout)

USB10G_CPU0_3
USB10G_P21_HUBUP
PCIE_ASM_X4
ASM_REFCLK
ASM_PERST_L
PCIE_I226A_X1
I226A_REFCLK
I226A_PERST_L
PCIE_I226B_X1
I226B_REFCLK
I226B_PERST_L
SUS_S3_N
SUS_S5_N
USB2_P21_HUB
ASM_CLKREQ_L
I226A_CLKREQ_L
I226B_CLKREQ_L
WAKE_L
BB_PWR_BTN_N
I2C_BB
SMB_ALERT_N

File: iob_hs.kicad_sch

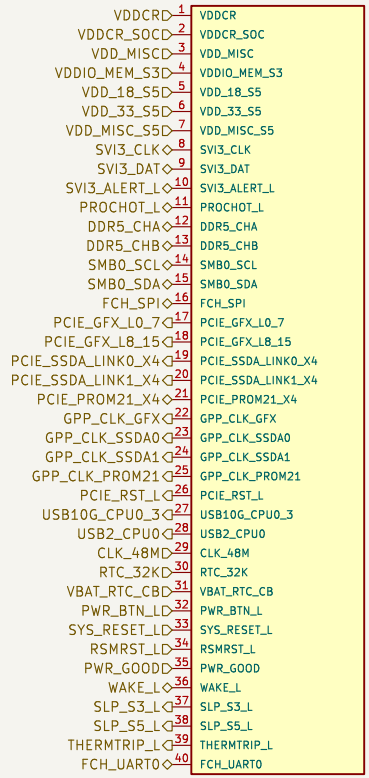
See /workspace/docs/am5-cpu-board-frozen-config.md; nets are block-level bundles
v4 frozen config: Ryzen 7000/9000 142 W, PROM21 PLACEHOLDER, TPS53685 SVI3, J3 rev2, no iGPU

MacPro6,2 / Aidan Winkler

Sheet: /
File: macpro62_am5.kicad_sch

Title: MacPro6,2 AM5 CPU board (CB) DRAFT — top-level block diagram

Size: A2	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 1/9



BLOCK STUB (rev A block diagram). Contents planned:

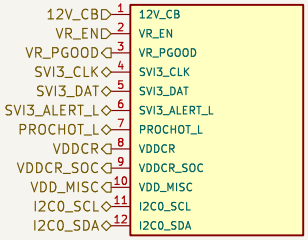
- U1 Socket AM5 LGA1718 (Foxconn PE17181/PE17186 or Lotes AZIFS055), MP62 contact frame on the 69.5 x 55 core holes
- Ryzen 7000/9000: 28 PCIe 5.0 lanes / 24 usable [AMD 9950X page; AMD Socket AM5 Quick Reference Guide Nov 2024]: 7000 per-SKU page [GAP-minor]
- Lane map (FROZEN): GFX x16 Gen5 -> J1 (GPU) | NVMe x4 Gen5 -> J1 (SSD board A link 0) | GPP x4 Gen5 -> J1 (SSD board A link 1) | x4 Gen4 -> PROM21 uplink
- SSD 'x8' = two separate x4 root ports, not one x8 (QRG split). GPP x4 Gen5 on every 7000 SKU [GAP]
- USB: 4 x native USB 3.2 Gen2 10G -> J3 C1-C4 (non-USB4) [AMD 9950X page]; 1 x native USB2 -> J1 USB2_SPARE
- iGPU display outputs REMOVED (iGPU unused per baseline)
- Pin names are functional placeholders: AM5 land map / pin list is AMD NDA [GAP]

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /CPU: Socket AM5 (Ryzen 7000/9000, 142 W PPT cap)/
File: cpu_am5.kicad_sch

Title: MP62 AM5 CPU board: CPU: Socket AM5 (Ryzen 7000/9000, 142 W PP

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 2/9



BLOCK STUB (rev A block diagram). Contents planned:

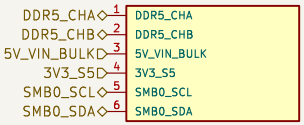
- U3 TI TPS53685 AMD SVI3 multiphase controller (public datasheet: ti.com/product/TPS53685) — replaces RAA229139
- Phase plan (unchanged estimate): VDDCR 5 phases, VDDCR_SOC 2, VDD_MISC 1 [rail/phase assignment vs TPS53685 channel limits: check datasheet]
- POWER-STAGE PAIRING TBD: SiC654 (current footprint) vs TI NexFET smart stages (e.g. CSD95xxx) — compatibility with TPS53685 current sense not yet checked
- Load line / IccMax / TDC / EDC from AMD AM5 VR guide (NDA) [GAP]; 142 W PPT -> VDDCR ~120 A over 5 phases [Estimate]
- VR_HOT_L -> PROCHOT_L; PMBus to EC (I2C0). Power limits managed in firmware (Aidan)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /SVI3 VRM: TI TPS53685 (VDDCR + VDDCR_SOC + VDD_MISC)/
File: vrm_svi3.kicad_sch

Title: MP62 AM5 CPU board: SVI3 VRM: TI TPS53685 (VDDCR + VDDCR_SOC +

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 3/9



BLOCK STUB (rev A block diagram). Contents planned:

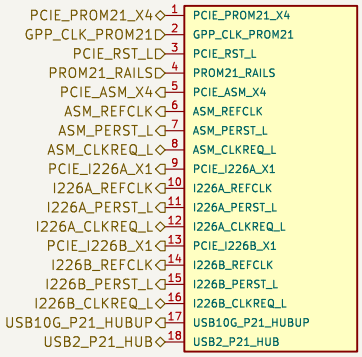
- J6/J7 (CH-A) and J9/J10 (CH-B): UMAX 90414 short-latch DDR5 UDIMM (unchanged)
- Module PMIC from 5V_VIN_BULK; SPD hub on SMBus0 (FCH), 3V3_S5; JEDEC UDIMM pin map not yet netted [open]

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /DDR5 2DPC: 4 x UDIMM vertical (back)/
File: ddr5.kicad_sch

Title: MP62 AM5 CPU board: DDR5 2DPC: 4 x UDIMM vertical (back)

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 4/9



BLOCK STUB (rev A block diagram). Contents planned:

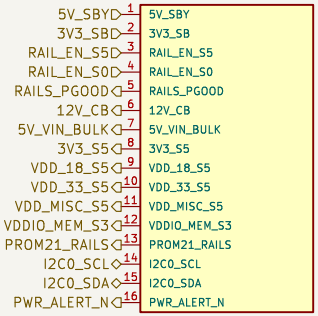
- *** PLACEHOLDER — pinout pending datasheet *** U2 AMD Promontory 21 (single chip, B650 class), FCBGA 19x19, ~7 W [press: Angstronomics, WePC; not an AMD datasheet]
- Ballout, power rails, straps, clocking: AMD NDA -> request via Communications only (no vendor contact) [GAP]
- Uplink: PCIe Gen4 x4 from CPU (PCIE_PROM21_X4, GPP_CLK_PROM21)
- Downlink #1 Gen4 x4 -> J3 -> ASM4242 USB4 (base side) | Downlink #2 Gen4 x4 SPARE (unrouted; OpenCore boot-drive home is open for Aidan)
- Flex x1 #1/#2 -> J3 -> i226-V #1/#2 (2.5 GbE); flex #3/#4 spare
- USB 10G -> J3 USB-A hub upstream (hub on base board); USB2 -> J3 base-board USB2 hub; other USB ports spare
- REFCLK outputs (ASM, i226A, i226B) + PERST#/CLKREQ# per port: [UNVERIFIED — PROM21 refclk outputs not in public data]
- Uplink oversubscription ~1.24x (relayed, not re-derived) [check]. Own SPI flash [Unverified]

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /PROM21 chipset — PLACEHOLDER — pinout pending datasheet/
File: prom21.kicad_sch

Title: MP62 AM5 CPU board: PROM21 chipset — PLACEHOLDER — pinout pend

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 5/9



BLOCK STUB (rev A block diagram). Contents planned:

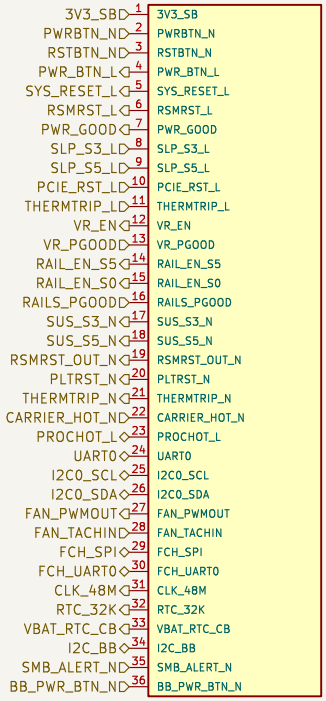
- LUG1/LUG2 12 V entry -> U11 TI TPS259851 eFuse (ILIM ~25 A) -> 12V_CB; U15 INA228 on I2C0, ALERT/FLT# -> PWR_ALERT_N
- !! 12 V PLANE MUST CARRY 17.9 A (142 W PPT, est.) / 26.3 A at 230 W - previously checked only to ~15 A -> RE-CHECK on JLC10161H-1080A (inner 0.5 oz) !!
- U4 CPU S5 rails from 5V_SBY / 3V3_SB [rail list Unverified, AMD NDA]; U8 VDDIO_MEM_S3; U7 5V_VIN_BULK (DIMM PMICs); U6 PROM21 rails [PLACEHOLDER, NDA]
- 3V3_S0 load switch; sequencing by EC (RAIL_EN_S5 / RAIL_EN_S0, RAILS_PGOOD)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /Platform power: 12 V entry (LUG1/LUG2), TPS259851 eFuse, aux + PROM21 rails/
File: power.kicad_sch

Title: MP62 AM5 CPU board: Platform power: 12 V entry (LUG1/LUG2), TPS25

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 6/9



BLOCK STUB (rev A block diagram). Contents planned:

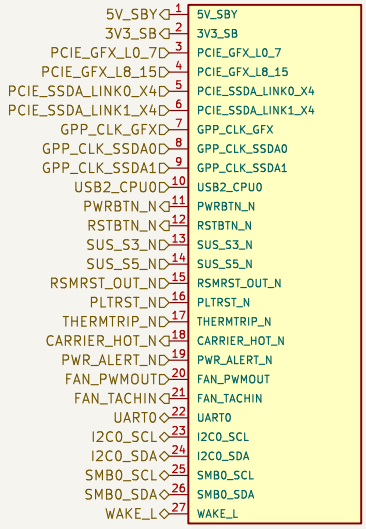
- U9 RP2350 EC: AM5 sequencing RSMRST_L -> PWR_BTN_L -> SLP_S5/S3 -> rails -> PWR_GOOD; VR_EN / VR_PGOOD
- Firmware: vendor AGESA UEFI for 7000/9000 via a licensed board maker (ODM) [NDA via ODM]; U10 FCH SPI ROM 32 MB; J4 TPM header (fTPM default)
- BT1 CR2032 ON-BOARD (holder footprint MP62_AREA_BT1_CR2032_22x16) -> diode-OR with 3V3_S5 -> VBAT_RTC_CB (VBAT no longer on J3)
- Y1 48 MHz [Unverified] + 32.768 kHz; J5 debug (FCH UART, SWD, POST); I2C to base MCU via J3 (LEDs/status)

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /EC, firmware, clocks, RTC (on-board CR2032), debug/
File: ec_fw.kicad_sch

Title: MP62 AM5 CPU board: EC, firmware, clocks, RTC (on-board CR2032), d

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 7/9



BLOCK STUB (rev A block diagram). Contents planned:

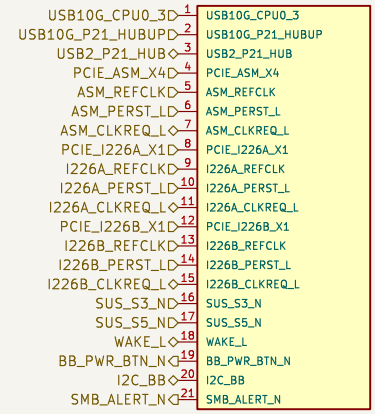
- J1 = Amphenol Mini Cool Edge 224 (ME1022410103011), 1.1 A/contact [Amphenol datasheet]; 78 GND (keep >= 78 in v4 map)
- Carries: GFX x16 Gen5 (GPU) + SSD board A link 0 x4 + link 1 x4 Gen5 + 3 REFCLKs; standby 6 x 5V_SBY + 2 x 3V3_SB; USB2 spare; sideband
- Contact-level v4 map (SSD A link0/link1 Gen5 replacing FS slot A/B Gen4, drop SATA0) not yet generated -> cpulink_224_pinout_am5_v4.csv [open]
- Gen5 redrivers (DS320PR1601 GPU, 2 x DS320PR810 SSD A) + PT5161L fallback live on the base board, not here

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /J1 Amphenol ME1022410103011 Mini Cool Edge 224 card edge/
File: cpu_link.kicad_sch

Title: MP62 AM5 CPU board: J1 Amphenol ME1022410103011 Mini Cool Edge

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 8/9



BLOCK STUB (rev A block diagram). Contents planned:

- J3 MCIO 124 RA (SFF-TA-1016) -> base board; contact map = docs/mp62-cb-j3_mcio124_host-end_am5_rev2.csv (67 signal + 57 GND, G-S-S-G)
- USB-C C1-C4 (non-USB4) = CPU native USB 10G x4 | USB-A hub upstream = PROM21 10G | USB2 -> base-board USB2 hub
- ASM4242 Gen4 x4 (PROM21 DL#1) + 2 x i226 x1 (PROM21 flex) + 3 REFCLKs [Unverified] + PERST#/CLKREQ#
- Sideband: SLP_S3#/S5#, 3 PERST#, 3 CLKREQ#, WAKE#, PWR_BTN#, I2C + SMB_ALERT#, BB_HS_PRSNT0#/1#
- Removed vs rev 1: iGPU DP (DDIB/DDIC/AUX/HPD), VBAT, USB_OC#, C5/C6, extra USB2

Block stub: hierarchical labels + one non-BOM stub symbol (#BLKn, passive pins) so ERC sees the n

Sheet: /J3 MCIO-124 to base board (rev 2 pinout)/
File: iob_hs.kicad_sch

Title: MP62 AM5 CPU board: J3 MCIO-124 to base board (rev 2 pinout)

Size: A4	Date: 2026-10-28	**Rev: A-am5-v4**
KiCad E.D.A. 9.0.2		Id: 9/9